



CMSC 440

Data Communication & Networking

Lec #3: Introduction to the Internet (III)



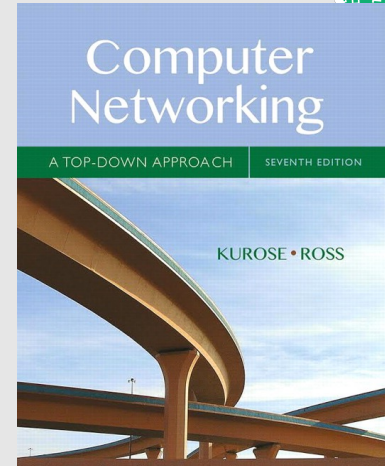
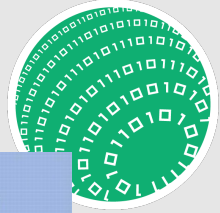
VCU

Tamer Nadeem

Introduction to the Internet

Overview

- What's the Internet (KR 1.1)
- Network Edge (KR 1.2)
- **Network Core (KR 1.3) (cont'd)**
- Performance: Loss and Delay (KR 1.4)
- Protocol Layering (KR 1.5)
- Networks Under Attack (KR 1.6)
- History (KR 1.7)



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Networking: A Top
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Jim Kurose, Keith Ross
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May 2016

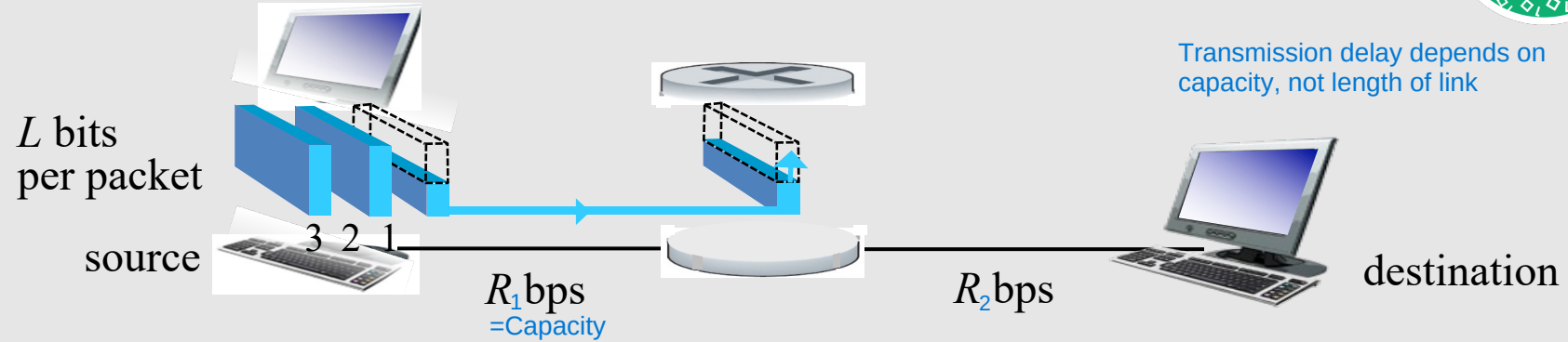
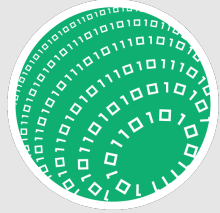
Packet Switching

Store and Forward

No dedicated paths

Packet 1 = $2L/R$

Every other = L/R



- Takes L/R seconds to transmit (push out) packet of L bits on to link at R bps
- **Store and forward**: entire packet must arrive at router before it can be transmitted on next link
- end-2-end delay (for one packet) = $2L/R$ (assuming 0 propagation delay)

- One-hop Example:

- $L = 7.5$ Mbits
- $R = 1.5$ Mbps

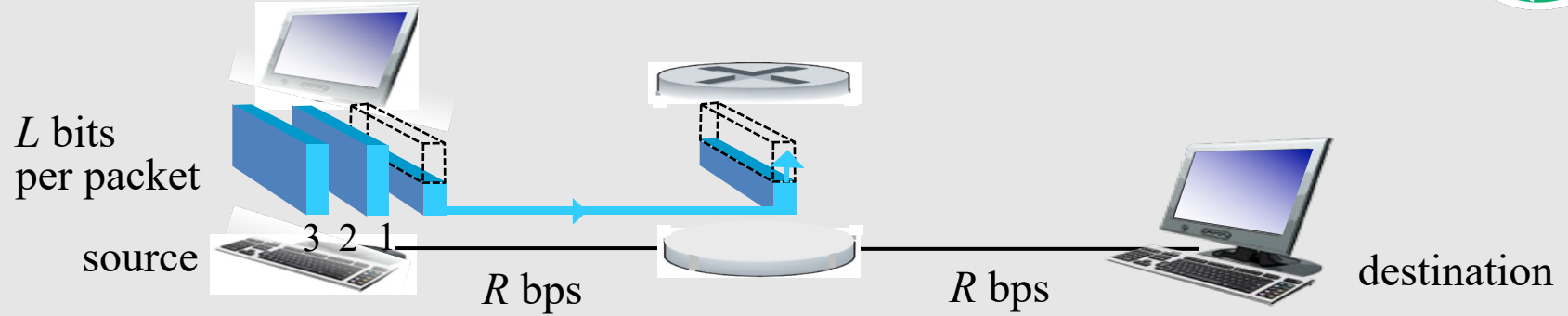
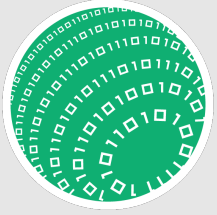
– one-hop transmission delay = 5 sec

$= L/R$ $= \text{Bandwidth}$

Transmission
Delay=first
performance metric

Store and Forward

Question

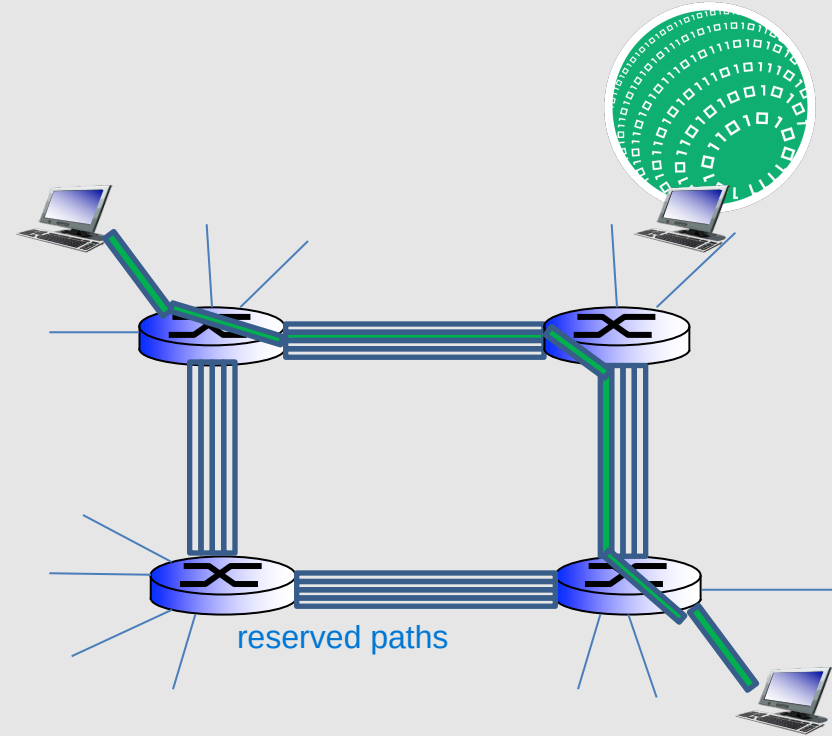


- How long until the destination receives all 3 packets?
- What is the delay for P packets of length L sent over a series of N links?

Network Core

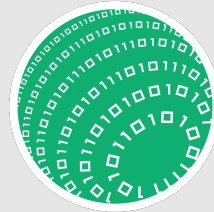
Alternative: circuit switching

- End-end resources allocated to, reserved for, "call" between source & destination
- In diagram, each link has four circuits.
 - call gets 2nd circuit in top link and 1st circuit in right link.
- Dedicated resources: no sharing
 - guaranteed performance
- Circuit segment idle if not used by call (**no sharing**)
- Commonly used in traditional telephone networks



Circuit Switching

FDM versus TDM



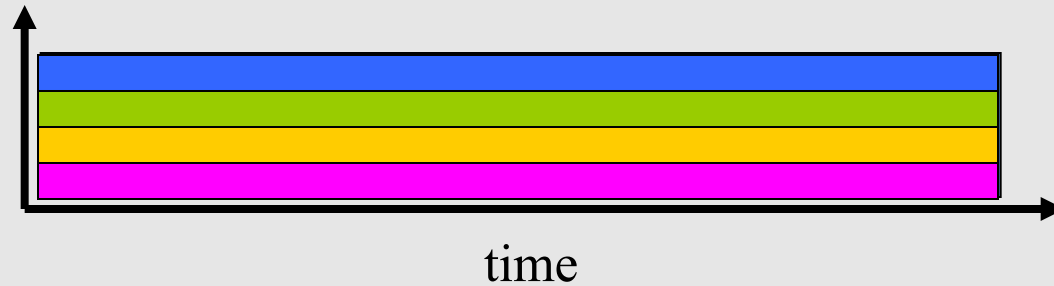
FDM (frequency division multiplexing)

Example:

4 users

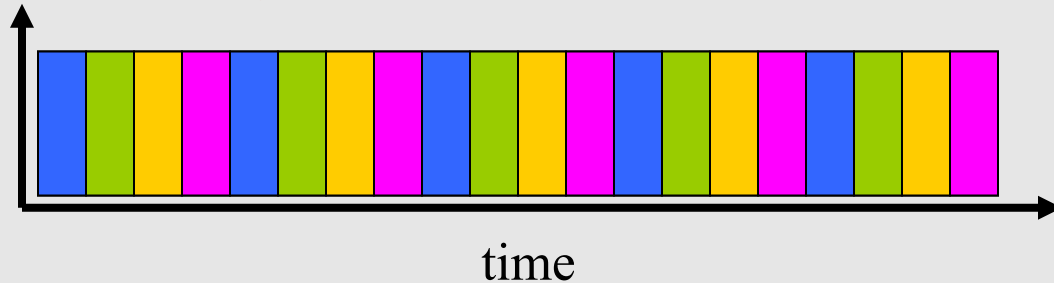


frequency



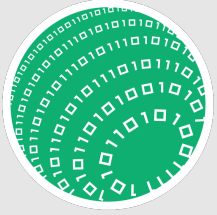
TDM (time division multiplexing)

frequency



Circuit Switching

Example



- How long would it take to send a file of 640,000 bits from Host A to Host B over a circuit-switched network?
 - All links have a total bitrate of 1.536 Mbps
 - Links use TDM with 24 slots
 - Takes 500 ms to setup a circuit

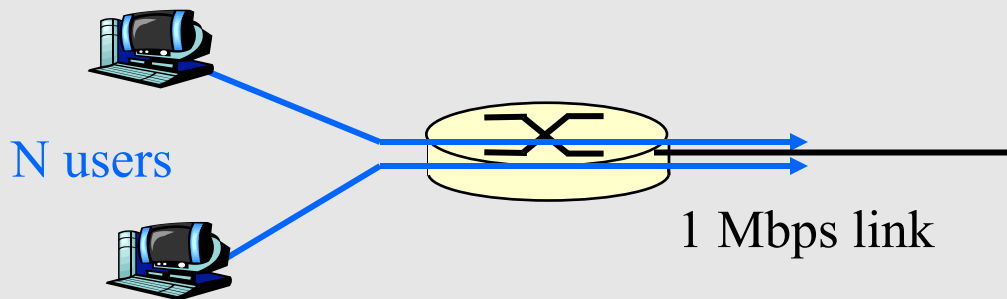
Packet Switching vs. Circuit Switching

Example



- 1 Mbps link
- each user:
 - 100 kbps when "active"
 - active 10% of time
- **circuit-switching:**
 - 10 users
- **packet switching:**
 - with 35 users, probability that more than 10 are active at same time is less than 0.0004

Packet switching allows more users to use network!

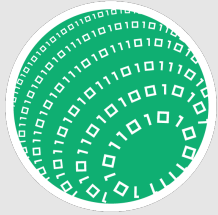


Q: how did we get value 0.0004?

Q: what happens if > 35 users ?

Packet Switching vs. Circuit Switching

Is packet switching a "no brainer"?

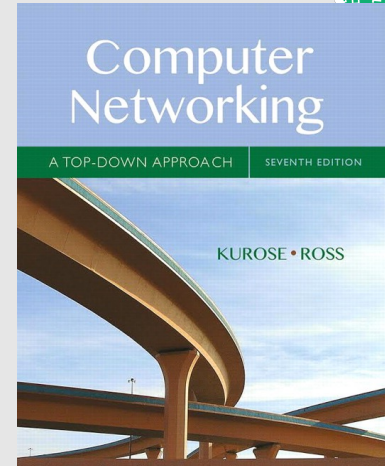


- Great for bursty data
 - Resource sharing
 - No call setup
- **Excessive congestion:**
packet delay and loss
 - Protocols needed for reliable data transfer, congestion control
- **How to provide circuit-like behavior?**
 - Bandwidth guarantees needed for audio/video applications?
 - Still an unsolved problem (that's why we do research!)

Introduction to the Internet

Overview

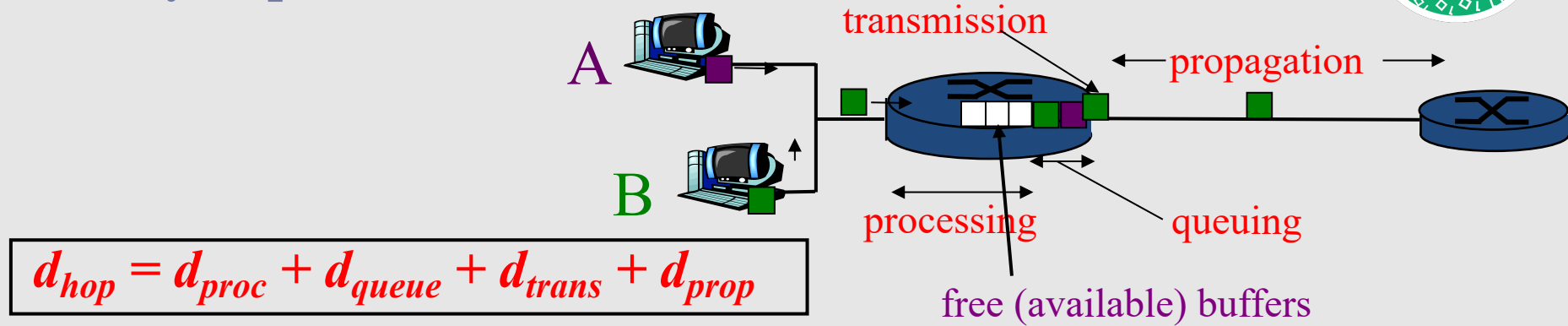
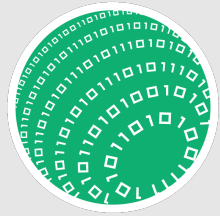
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Performance: Loss and Delay

Delay in packet-switched networks



Four sources of delay at each hop

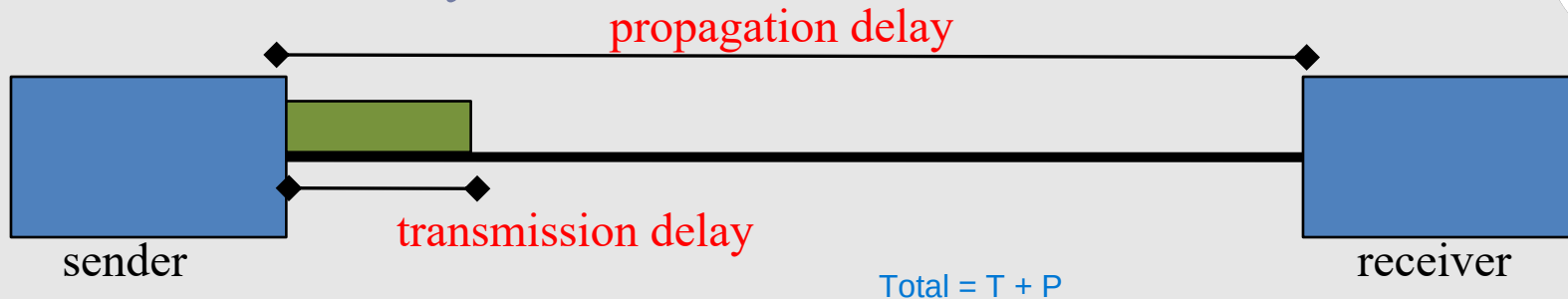
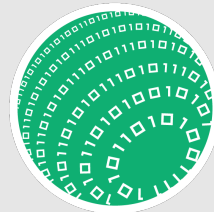
- ▶ Packets experience variable delays along the path from source to destination
- **Transmission Delay (d_{tran})**
 - time to "put bits onto the link" = L/R
 - R = link bandwidth (bps)
 - L = packet length (bits)
- **Propagation Delay (d_{prop})**
 - time to "traverse the link" = d/s
 - d = length of physical link
 - s = signal propagation speed in medium ($\sim 2 \times 10^8$ m/s)

Check out the online interactive exercises for more examples: http://gaia.cs.umass.edu/kurose_ross/interactive/

See Java applet: https://media.pearsoncmg.com/aw/ecs_kurose_compnetwork_7/cw/content/interactiveanimations/transmission-vs-propagation-delay/transmission-propagation-delay-ch1/index.html

Delay in Packet-Switched Networks

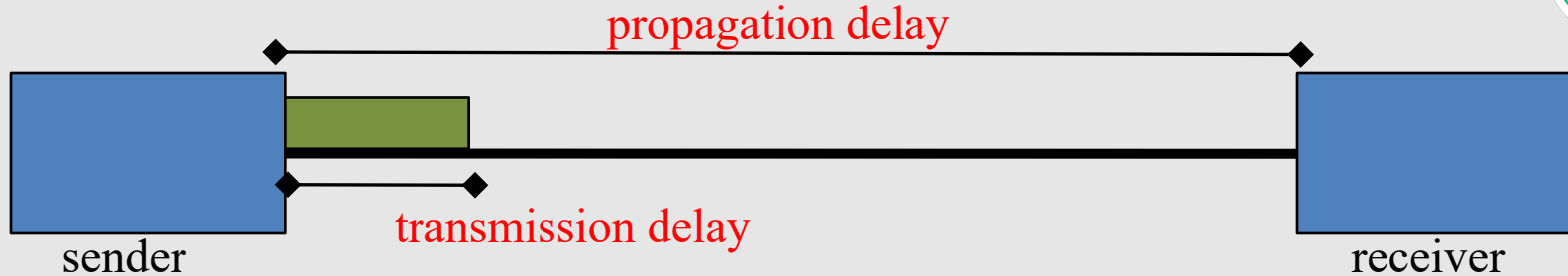
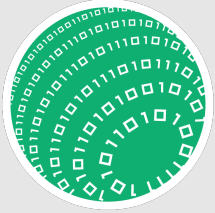
Transmission Delay



- **Transmission Delay**
 - time it takes to put (transmit) a *packet* on the wire
 - depends on the length (size) of the **packet** and the **transmission speed** (or link speed or bandwidth)
 - tells you when the last bit of the packet is transmitted (leave the sender)
- ▶ **Calculation** ($d_{trans} = L / R$)
 - ▶ divide the size of the packet (L) by the transmission speed (R)
 - ▶ end result is a time, so units should be seconds or milliseconds

Delay in Packet-Switched Networks

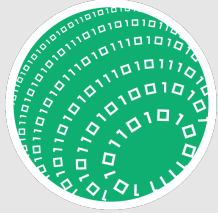
Propagation Delay



- **Propagation Delay**
 - time a *bit* takes to travel (propagate) the length of a wire
 - depends upon the **length** of the wire and the **propagation speed** of the physical medium (coaxial cable, fiber optics, air, etc.)
 - tells you when the first bit will reach the other end of the wire
- ▶ **Calculation** ($d_{prop} = m / s$)
 - ▶ divide the length of the wire (m) by the propagation speed (s) (typically 2.5×10^8 m/s)
 - ▶ end result is a time, so units should be seconds or milliseconds

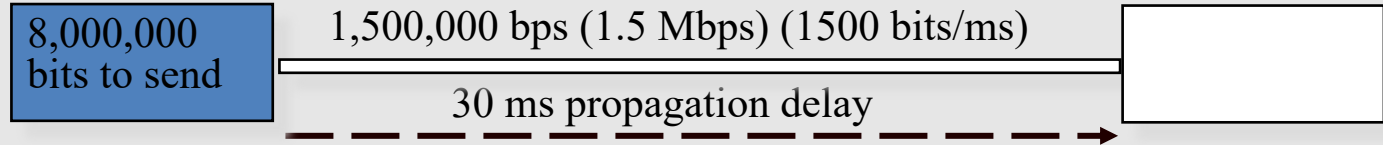
Transmission & Propagation Example

Transmission on a "slow" link



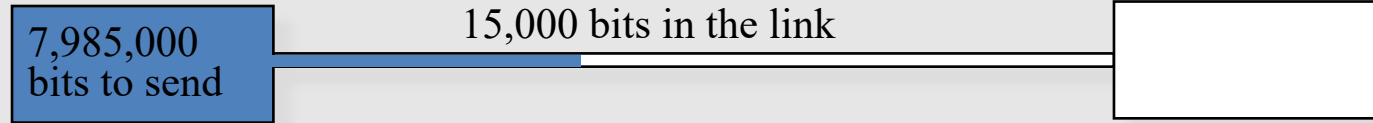
R

Time = 0



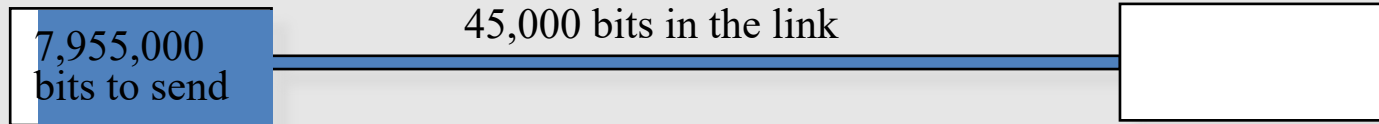
Time = 10 ms

Trans

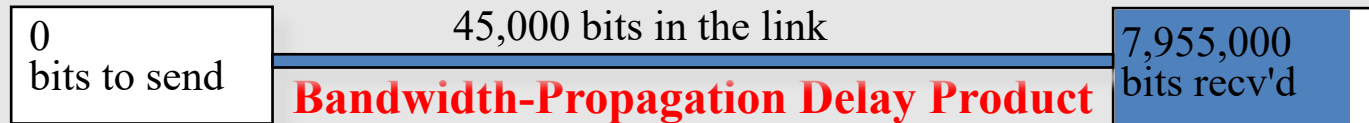


Time = 30 ms

Prop



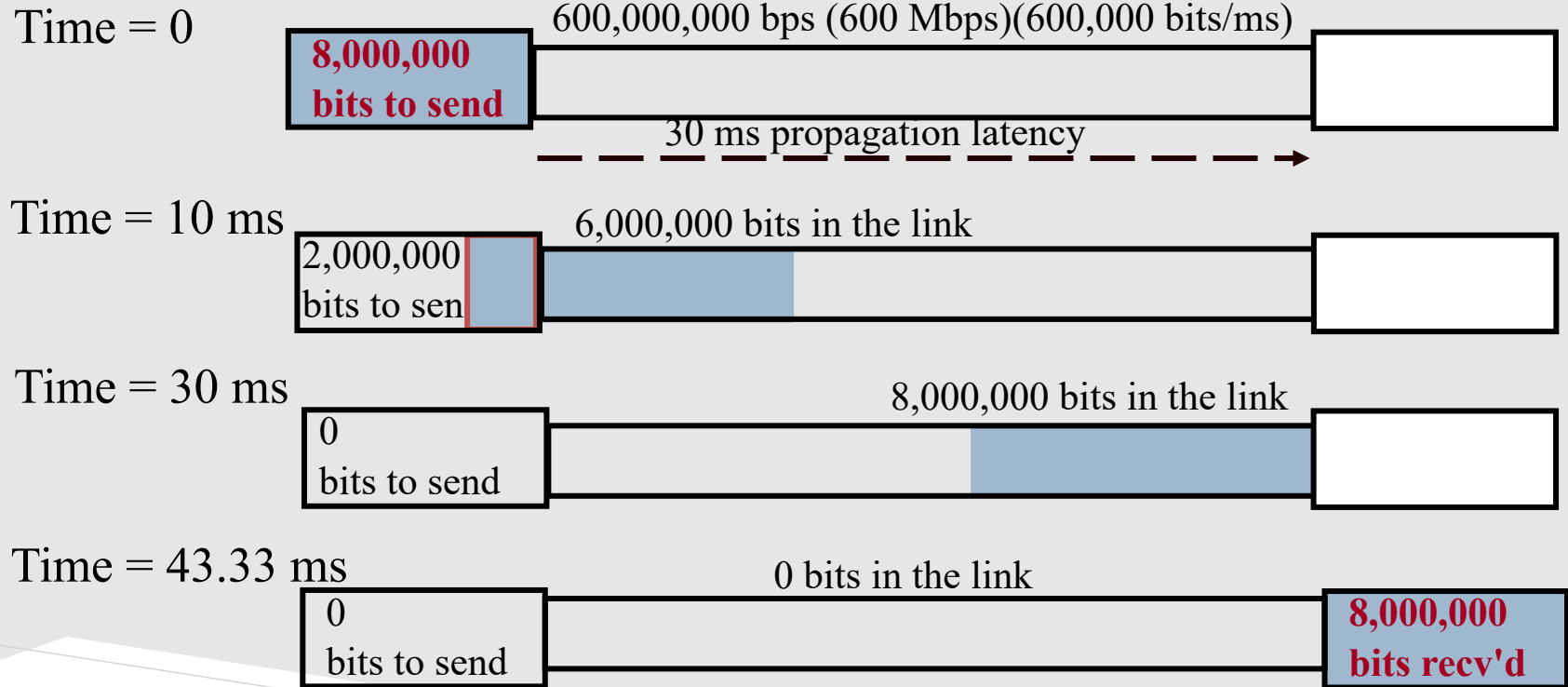
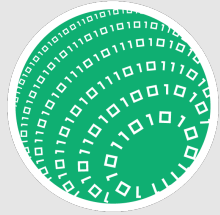
Time = 5333.3 ms



**Bandwidth-Propagation Delay Product
(BDP)**

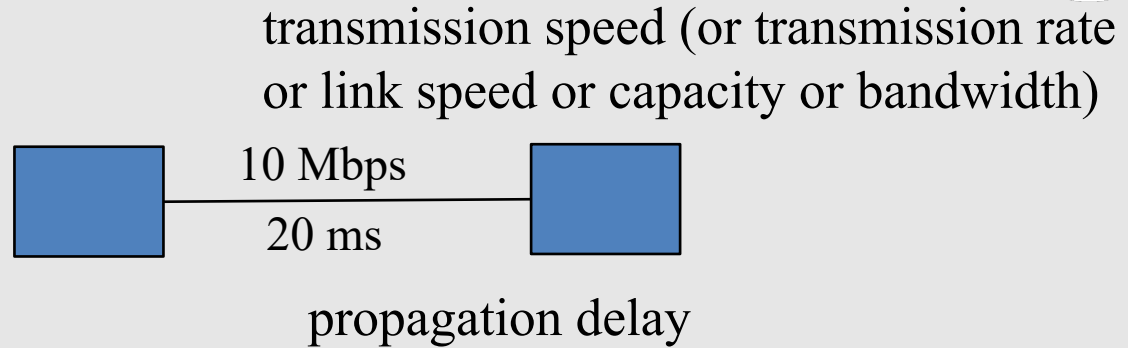
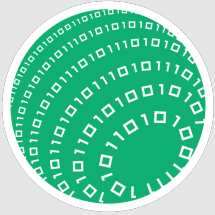
Transmission & Propagation Example

Transmission on a "fast" link



Delay in Packet-Switched Networks

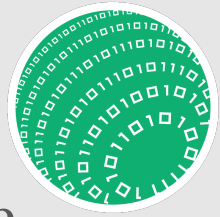
Transmission & Propagation Delay



- end-to-end delay
 - time it takes for the entire packet to reach the receiver (i.e., when does the last bit of the packet reach the receiver?)
 - transmission delay + propagation delay + processing delay (if applicable) + queuing delay (if applicable)

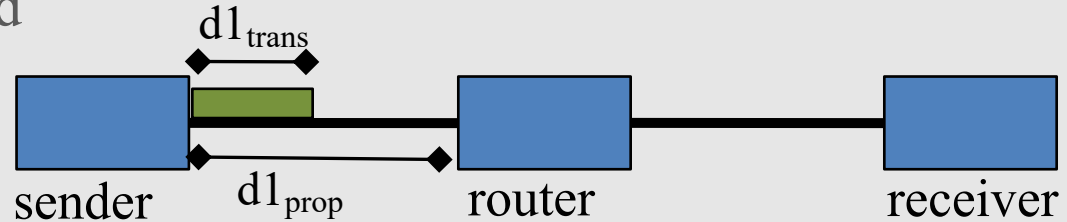
Delay in Packet-Switched Networks

Effect of Store and Forward

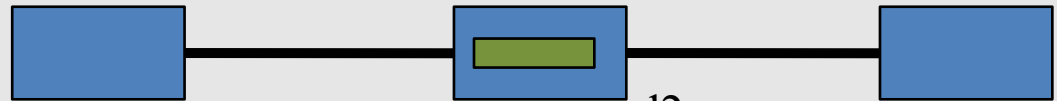


- Can't begin transmitting a packet over the next link until the entire packet has first been received

$$\text{time} = d1_{\text{trans}}$$



$$\text{time} = d1_{\text{trans}} + d1_{\text{prop}}$$



$$\text{time} = d1_{\text{trans}} + d1_{\text{prop}} + d2_{\text{trans}}$$

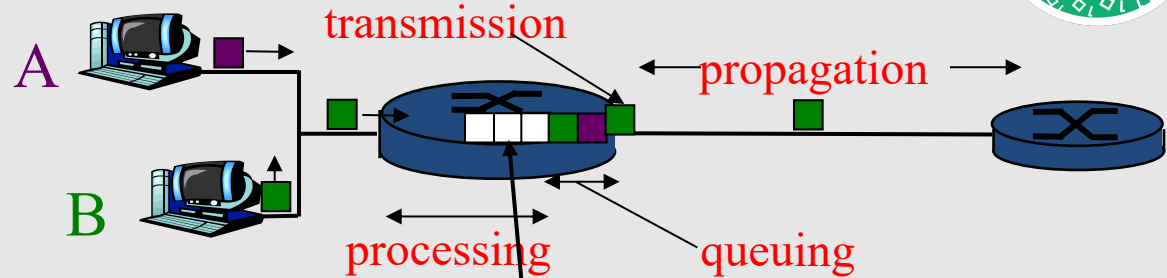


$$\text{time} = d1_{\text{trans}} + d1_{\text{prop}} + d2_{\text{trans}} + d2_{\text{prop}}$$



Performance: Loss and Delay

Delay in packet-switched networks



$$d_{hop} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

free (available) buffers: arriving packets dropped (lost) if no free buffers

– Processing Delay (d_{proc})

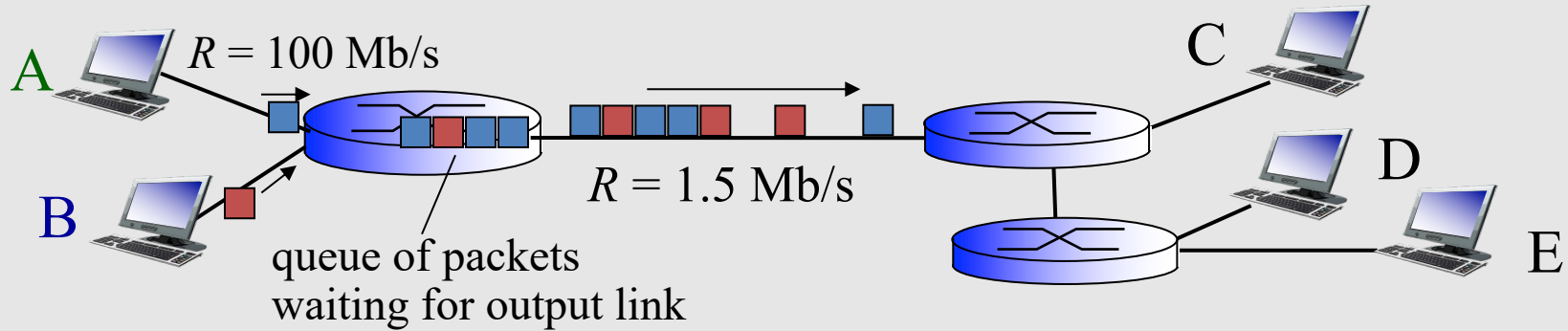
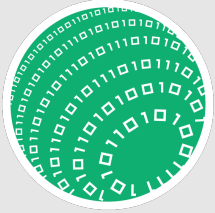
- Check for bit errors
- Determine the output interface to forward packet on

– Queuing Delay (d_{queue})

- Time spent waiting at outbound interface for transmission
- Duration depends on the level of congestion at the interface

Packet Switching

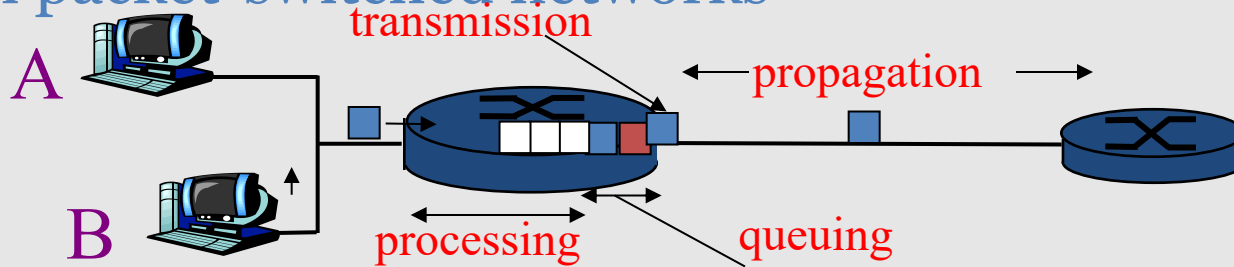
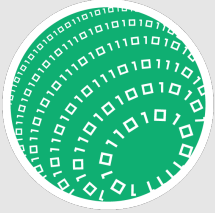
Queuing delay, loss



- If arrival rate (in bits) to link exceeds transmission rate (capacity) of link for a period of time:
 - packets will queue, wait to be transmitted on link
 - packets can be dropped (lost) if memory (buffer) fills up

Performance: Loss and Delay

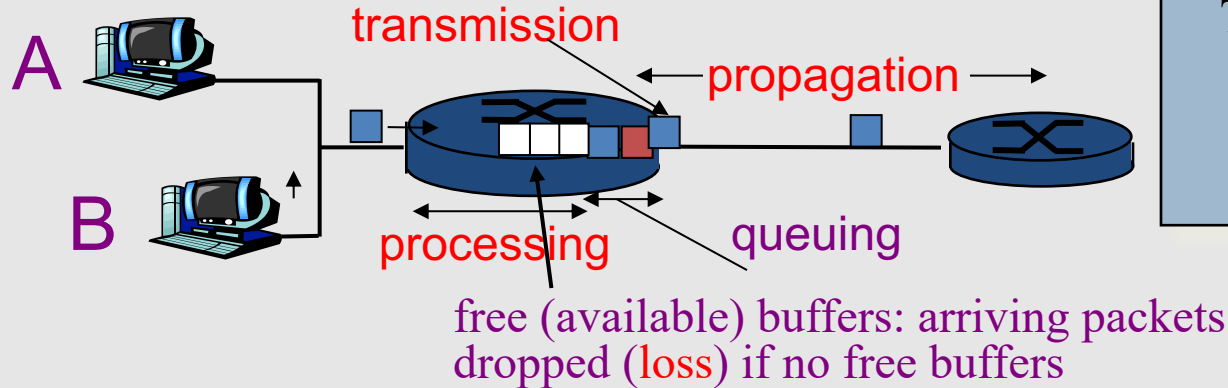
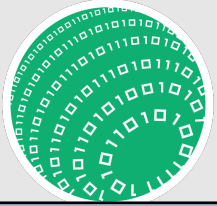
Delay in packet-switched networks



- **Typical transmission delay:**
 - L/R (significant for low-speed links)
 - $120 \mu\text{s}$ (for 1,500 byte packet on a 100 Mbps Ethernet)
- **Typical processing delay:**
 - a few micro-secs or less
- **Typical queuing delay:**
 - depends on congestion: *packet arrival rate* to link (temporarily) exceeds output link capacity
- **Typical propagation delay:**
 - $\leq 1 \mu\text{s}$ on a small campus
 - $\approx 25 \text{ ms}$ to the West coast

Performance: Loss and Delay

Delay in packet-switched networks



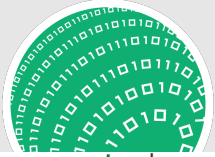
Transmission delay = $120 \mu\text{s}$
Propagation delay = $10 \mu\text{s}$
Processing delay = $1 \mu\text{s}$
Queuing delay = $k \times 120 \mu\text{s}$

What is k ?

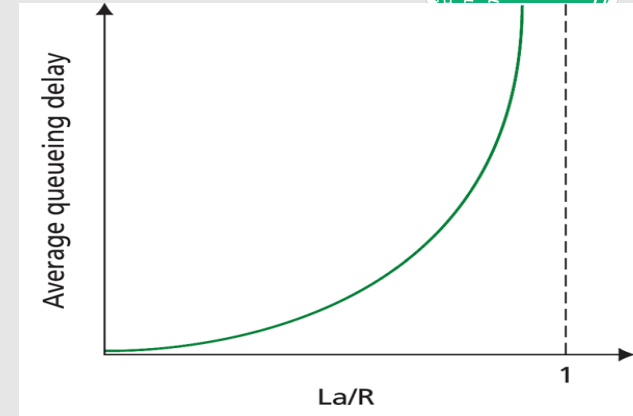
- ▶ What dominates end-to-end delay?
- ▶ Note that processing, transmission, and queuing delays are encountered at each hop
 - ▶ End-to-end delay is largely a function of the number of routers encountered along the path from source to destination

Delay in Packet-Switched Networks

Queuing Delay

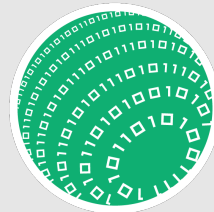


- What affects queuing delay?
 - traffic arrival rate (La)
 - Packet length in bits (L), average packet arrival rate (a)
 - speed of outgoing link (R)
 - nature of arriving traffic (uniform or bursty)
 - hard to quantify
- Represent queuing delay with statistical measures
 - average queuing delay
 - variance of queuing delay
 - Probability queuing delay exceeds some value
- Traffic intensity (La/R)
 - $La/R \sim 0$ - avg queuing delay small
 - $La/R \rightarrow 1$ - avg queuing delay large
 - $La/R > 1$ - more "work" arriving than can be serviced, avg delay infinite!

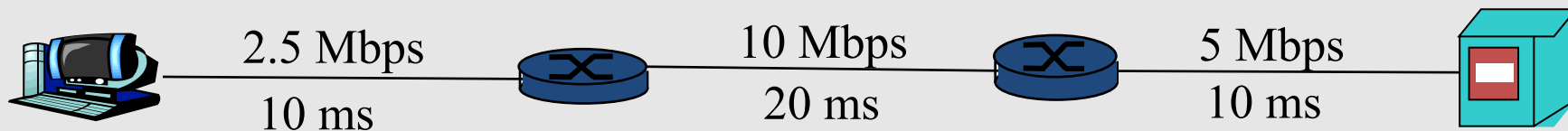


See Java applet: https://media.pearsoncmg.com/aw/ecs_kurose_compnetwork_7/cw/content/interactiveanimations/queuing-loss-applet/index.html

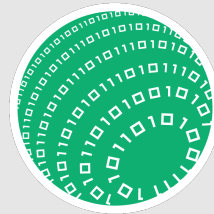
Question



- How long would it take a 5000-bit packet to travel from source to destination over the following network?



Questions



- What is the transmission delay for a 2000-byte packet over a 1 Mbps link?
- Where will be the last bit of the packet after the transmission delay has passed?
- What is the transmission delay for a 250,000-bit packet over a 1 Mbps link?

Questions



- What are the four types of delay that are (potentially) encountered at every router/hop? Which one may not be encountered?
- What is the difference between transmission delay and propagation delay?
- What two things affect the duration of queuing delay for a particular packet?



Questions?

